

2018-Ph-H1-Q25

Section: Electricity

Topic: Monitoring and Measuring a.c.

An oscilloscope shows 4 divisions between peaks. Timebase = 5 ms/div.

What is the frequency?

Solution:

Period:

$$T = 4 \times 5 = 20 \text{ ms} = 0.020 \text{ s}$$

Frequency:

$$f = \frac{1}{T} = \frac{1}{0.020} = 50 \text{ Hz}$$

Answer: 50 Hz

Revision Tip:

Timebase $\times$ divisions gives period.

Then $f = 1/T$. Always convert ms to s!
